

Enhancing Dog Breed Classification with Generative Data Augmentation

Group Members

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Git Repository

<https://github.com/IsuruAriyaratne97/genVision>

Objective

The objective of this project is to enhance the performance of a multi-class image classifier designed to distinguish between various dog breeds. To achieve this, we aim to augment the training dataset using generative models such as GANs (Generative Adversarial Networks) and diffusion models. The effectiveness of these synthetic augmentations will be measured by evaluating the classifier's performance before and after the inclusion of generated images. The central hypothesis is that the additional data will improve classification accuracy and robustness by increasing the diversity of training samples, particularly for underrepresented classes.

Dataset

The dataset used in this project comprises images of dogs, from “Stanford Dogs Dataset”. These images represent multiple dog breeds.

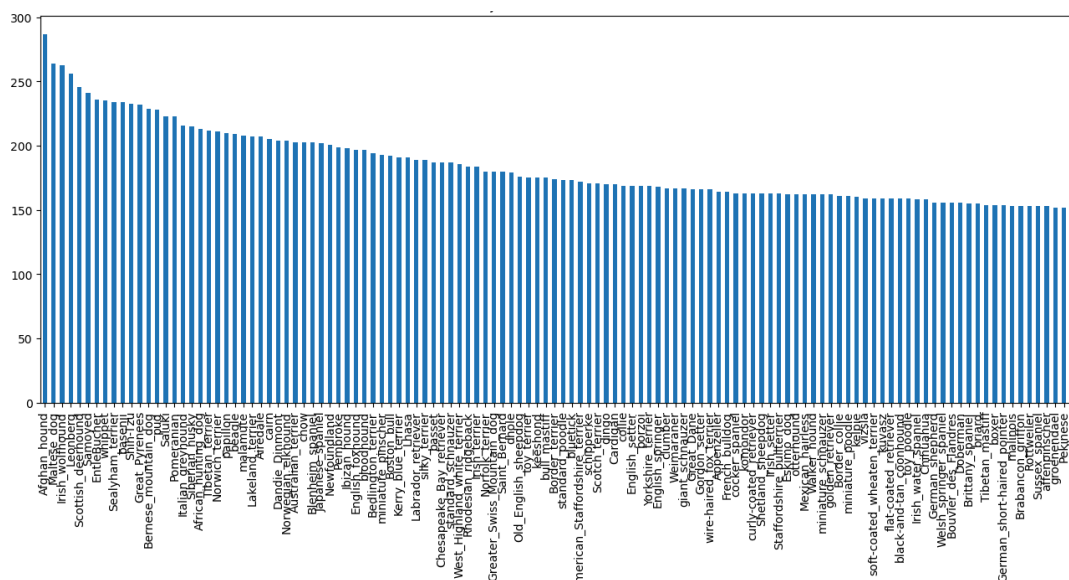


Figure1 : Class Distribution

Dataset

<https://www.kaggle.com/competitions/generative-dog-images/discussion?sort=hotness>

Experimental Setup

The experiment will be conducted in three phases.

Phase 1 (Baseline) - A classifier will be trained and evaluated using only the original dataset.

- **Model Architecture:**
 - CNN-based model (e.g., ResNet, EfficientNet, or a custom architecture)
- **Loss Function:**
 - Categorical Cross-Entropy
- **Training Strategy:**
 - Early stopping to avoid overfitting
 - Learning rate scheduler for optimized convergence
 - 80/20 train-validation split
- **Evaluation Metrics:**
 - Overall Accuracy
 - Per-class Precision and Recall
 - Confusion Matrix

Phase 2 (Generation) - Synthetic images will be generated for each class using the selected generative models.

- **GANs (Generative Adversarial Networks):**
 - Conditional GAN (cGAN) or StyleGAN
 - **Challenges:**
 - Mode collapse
 - Training instability
 - Ensuring class-conditional image fidelity
- **Diffusion Models:**
 - Denoising Diffusion Probabilistic Models (DDPM) or Stable Diffusion fine-tuned on the dataset
 - **Advantages:**
 - High-quality and diverse image generation
 - Greater training stability than GANs

Phase 3 (Augmented Training) - Synthetic images will be combined with the real training data. Then using the combined dataset the classifier will be retrained. We will then compare its performance against the baseline to assess the impact of the augmentation.

Expected Outcome

We anticipate that augmenting the dataset with high quality synthetic images will lead to improved accuracy, especially for dog breeds that are underrepresented in the original dataset. The use of generative models should help increase intra-class diversity and introduce novel feature variations that support better generalization.